Problem-Solution Fit

Date	24 July 2026
Team ID	SWUID2026211863
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:
Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Farmers (Small & Large Scale) • Agricultural Researchers • Agribusiness Companies • Government & Agricultural Officers	**6. CUSTOMER LIMITATIONS** EG. BUDGET, DEVICES **[CL]** • Limited internet connectivity • Limited technical knowledge • Budget constraints • Limited access to agricultural experts	**5. AVAILABLE SOLUTIONS** PROS & CONS **[AS]** **Solutions:** Traditional farming advice, agricultural experts, online farming websites. **Pros:** Easy access, experienced guidance. **Cons:** Time-consuming, expensive, generalized recommendations, not always data-driven.
2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]** • Difficulty selecting the right crop for soil conditions (Very High) • Low crop yield due to poor decisions (High) • Lack of scientific farming guidance (High) • Inefficient use of fertilizers and water (Medium)	**9. PROBLEM ROOT / CAUSE [RC]** • Lack of data-driven crop selection. • Changing weather conditions. • Poor understanding of soil nutrients. • Limited access to precision agriculture technologies.	**7. BEHAVIOR** + ITS INTENSITY **[BE]** • Frequently consults local farmers and experts. • Uses weather forecasts before planting. • Searches online for crop recommendations. **Intensity:** High during every cultivation season.

3. TRIGGERS TO ACT [TR] • Beginning of a new farming season • Soil test results received • Need to improve crop productivity • Climate and weather changes	**10. YOUR SOLUTION [SL]** **OptiCrop** is a machine learning-based crop recommendation system that analyzes Nitrogen, Phosphorous, Potassium, Temperature, Humidity, pH, and Rainfall to recommend the most suitable crop through a simple Flask web application, enabling datadriven and sustainable farming decisions.	**8. CHANNELS OF BEHAVIOR [CH]** **ONLINE :** Agricultural websites, government portals, YouTube, mobile apps.
4. EMOTIONS BEFORE / AFTER **[EM]** **Before:** Confused, uncertain, worried about crop failure. **After:** Confident, informed, satisfied, optimistic about better yield.		**OFFLINE :** Agricultural officers, local markets, fellow farmers, training programs.